

Math 140H — Honors Calculus I: Lecture Notes

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Welcome!

These notes accompany Math 140H, Honors Calculus I, Fall 2026. Each lecture here corresponds to one class session and covers more or less the material presented in class, in the same order, so that it can serve as a compact study reference. HTML “hardcopies” of the Jupyter notebook activities from class are also included. Lecture notes will generally be available immediately **after** the corresponding class session.

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Part I

Real Numbers

1 The Rational Numbers and the Number Line

Companion reading

This lecture corresponds to C&J §1.1a, together with the neighborhood definition from §1.1d — covered a bit early here since we will use it constantly for the rest of the semester.

Calculus is built on the real numbers, so before we can talk about limits, derivatives, or integrals we need to understand exactly what a real number *is* — and, just as importantly, why the rational numbers alone are not enough. This lecture makes that precise: we recall how the natural numbers extend to the rationals, set up the number line, and then prove that a length as ordinary as the hypotenuse of a simple right triangle cannot be a rational number at all.

1.1 From Counting to Measuring

There are two major uses for numbers: **counting** and **measuring**. The natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ are the tool for counting — abstract symbols for how many elements are in a collection, stripped of any reference to what is actually being counted. They come equipped with addition and multiplication satisfying a short list of familiar rules.

Remark 1.1. For all natural numbers a, b, c :

- $a + b = b + a$ and $ab = ba$ (commutative laws)
- $a + (b + c) = (a + b) + c$ and $a(bc) = (ab)c$ (associative laws)
- $a(b + c) = ab + ac$ (distributive law)
- $a + c = b + c \Rightarrow a = b$ (cancellation law)

These are the *rules* of arithmetic, not its *content* — everything else in this course is built on top of them, and we will not dwell on them further.

Natural numbers can be added and multiplied, but not always subtracted or divided. There is no natural number a with

$$5 + a = 2,$$

so subtraction is not always possible in $\mathbb{N}$. Extending $\mathbb{N}$ to the **integers** $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ fixes this: now $5 + (-3) = 2$.

In $\mathbb{Z}$, we still cannot always divide: there is no integer $b \in \mathbb{Z}$ with

$$5 \cdot b = 2.$$

Extending $\mathbb{Z}$ to the **rational numbers** fixes this in turn.

Definition 1.1.

A **rational number** is a number that can be written as p/q where $p, q \in \mathbb{Z}$ and $q \neq 0$. Every rational number has a unique representation p/q with $q > 0$ and $\gcd(p, q) = 1$ (no common factor larger than 1); we call this representation **in lowest terms**.

Within $\mathbb{Q}$, all four rational operations — addition, subtraction, multiplication, and division (except by 0) — can be carried out freely, and they obey exactly the same laws as in Remark 1.1. In this precise sense, $\mathbb{Q}$ is the smallest extension of the natural numbers in which counting-type arithmetic becomes measuring-type arithmetic.

1.2 The Number Line

To turn numbers into geometry, fix a line L , a point O on it called the **origin**, and a second point I to the right of O ; the segment OI serves as our unit scale for measuring — a “prototype meter.” The direction from O to I is called **positive**.

Every point on L is now uniquely determined by two pieces of information: its **distance** from O , and its **direction** (left or right) from O . For example, the point at distance $\frac{1}{2}$ in the positive direction represents the rational number $\frac{1}{2}$; the point at distance 2 in the positive direction represents 2.

Definition 1.2.

The distance of a number x from the origin is written $|x|$ and called the **absolute value** of x :

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0. \end{cases}$$

Note $|x| \geq 0$ always, with equality only when $x = 0$.

Every rational point can in fact be *constructed*, not just described: to locate p/q on L with ruler and compass, construct a subdivision of the segment OI into q equal parts, then extend the part closest to O , p times. We will identify each rational number with its representation on L .

1.3 Distances and Intervals

For $x, y \in L$, the number $|x - y|$ is the **distance between x and y** . Observe: if $x, y \in \mathbb{Q}$, then $|x - y| \in \mathbb{Q}$ as well — distances between rational points are themselves rational.

Definition 1.3.

For $a < b$, the **interval** between a and b comes in four flavors:

- open: $(a, b) := \{x : a < x < b\}$ (both endpoints excluded)
- closed: $[a, b] := \{x : a \leq x \leq b\}$ (both endpoints included)
- half-open: $[a, b) := \{x : a \leq x < b\}$ and $(a, b] := \{x : a < x \leq b\}$

Definition 1.4.

Given a number a and $\varepsilon > 0$, the **ε -neighborhood of a** is the open interval

$$\{y : |y - a| < \varepsilon\} = (a - \varepsilon, a + \varepsilon),$$

i.e. the set of points less than ε away from a .

We will meet ε -neighborhoods constantly once we get to limits — keep this definition close by.

1.3.1 Density

Proposition 1.1.

Between any two distinct rational numbers $a < b$ there is another rational number c with $a < c < b$. Consequently, between a and b there are infinitely many rational numbers.

Proof. Let $c = \frac{a+b}{2}$. Since $a, b \in \mathbb{Q}$ and $\mathbb{Q}$ is closed under addition and division by 2, $c \in \mathbb{Q}$; and since $a < b$, averaging gives $a < c < b$.

Now repeat the argument on the pair a, c to get a rational number strictly between a and c , hence strictly between a and b . Doing this again and again produces infinitely many rational numbers between a and b . $\square$

We say the rationals are **dense** on the number line: no matter how closely you look, you will always find another one.

1.4 Incommensurable Lengths

Given how constructible and how dense $\mathbb{Q}$ is, a natural question arises: is there anything “left in between” the rational numbers? Are there any *gaps*? Clearly, such a gap could not itself be an interval — density rules that out, since any interval of positive length contains infinitely many rational points. If a gap exists at all, it can only be a single missing point.

Remark 1.2. The discovery that not every length is a rational multiple of a chosen unit is due to the school of Pythagoras, in the fifth or sixth century BCE. Consider a right triangle with both legs of length 1; by the Pythagorean theorem, its hypotenuse a satisfies

$$1^2 + 1^2 = a^2 = 2,$$

and a can certainly be constructed with ruler and compass. The theorem below shows this length cannot be written as p/q for integers p, q — the hypotenuse and the legs of this triangle are, in the language of the ancients, **incommensurable**: there is no common unit of which both are integer multiples.

Theorem 1.1.

There is no rational number x with $x^2 = 2$. Equivalently, $\sqrt{2} \notin \mathbb{Q}$.

Proof. Suppose, toward a contradiction, that $\sqrt{2} = p/q$ for integers p, q with $q \neq 0$ and $\gcd(p, q) = 1$ (i.e. p/q is in lowest terms).

1. Square both sides and clear the denominator: $p^2 = 2q^2$.
2. Conclude that p^2 is even — if p were odd, p^2 would be odd, but $p^2 = 2q^2$ is even. Write $p = 2r$ for some integer r .
3. Substitute $p = 2r$ into the equation from Step 1 and simplify to get an equation for q^2 in terms of r : $(2r)^2 = 2q^2 \Rightarrow 4r^2 = 2q^2 \Rightarrow q^2 = 2r^2$.
4. Conclude that q is also even, by the same reasoning as in Step 2.
5. Steps 2 and 4 show that 2 divides both p and q . This contradicts $\gcd(p, q) = 1$, which we assumed when we wrote p/q in lowest terms.
6. Conclude: $\sqrt{2}$ is irrational — no integers p, q satisfy $\sqrt{2} = p/q$, so $\sqrt{2} \notin \mathbb{Q}$.

□

Exercise 1.1. Adapt the proof of Theorem 1.1 to show that $\sqrt{3}$ is irrational. Then try to see where the argument would break if you attempted the same proof for $\sqrt{4}$ — since $\sqrt{4} = 2$ is certainly rational, something in the argument *must* fail. Where, exactly?

1.5 Summary

- The rationals $\mathbb{Q}$ satisfy all the usual laws of arithmetic (Remark 1.1) and are **dense**: between any two rationals lies another (Proposition 1.1).
- Distances $|x - y|$, intervals, and ε -neighborhoods (Definition 1.3, Definition 1.4) give us the basic vocabulary for talking about “closeness” on the number line — vocabulary we will lean on heavily once limits arrive.
- Density notwithstanding, $\mathbb{Q}$ does not exhaust the number line: Theorem 1.1 exhibits an ordinary, constructible length that is not a rational number.
- The rationals therefore leave *gaps* in the line. Lecture 2 closes these gaps rigorously, using **nested intervals** to construct the real numbers.

2 The Real Numbers via Nested Intervals

Companion reading

This lecture corresponds to C&J §1.1b–c.

Last time we found a gap: the length $\sqrt{2}$ is constructible with ruler and compass, yet Theorem 1.1 shows it is not a rational number. So what *is* $\sqrt{2}$, as a number? We are used to describing such quantities through infinite decimal expansions, e.g.

$$\pi = 3.1415926 \dots,$$

but so far this is just a string of digits — it is not clear what it *means* mathematically. The key idea of this lecture is that an infinite decimal (or any other description of an irrational number) is really a recipe for **approximating** a point by rational numbers, to any desired degree of accuracy. We make this precise using shrinking intervals with rational endpoints.

2.1 Nested Intervals

Suppose x is a point on the number line that we want to pin down using only rational numbers. Start with any closed interval $I_1 = [a_1, b_1]$ with $a_1, b_1 \in \mathbb{Q}$ and $a_1 \leq x \leq b_1$. Now choose a subinterval $I_2 = [a_2, b_2] \subseteq I_1$ that still contains x :

$$a_1 \leq a_2 \leq x \leq b_2 \leq b_1,$$

for instance by splitting I_1 in half and keeping whichever half contains x . Repeating this forever produces a chain

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots, \quad x \in I_n \text{ for every } n.$$

For this to genuinely *pin down* x — rather than just hover near it — we need one more ingredient: the intervals must shrink all the way down to a point.

Definition 2.1.

A **nested sequence of closed intervals with rational endpoints** is a sequence $I_n = [a_n, b_n]$, $n = 1, 2, 3, \dots$, with $a_n, b_n \in \mathbb{Q}$, such that

1. $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$, and
2. the lengths $|I_n| := b_n - a_n$ tend to 0 as $n \rightarrow \infty$: for every $\varepsilon > 0$ there is an N such that $|I_n| < \varepsilon$ for all $n \geq N$.

Proposition 2.1.

If a nested sequence of closed intervals (I_n) contains a point in every I_n , that point is unique: there is at most one x with $x \in I_n$ for all n .

Proof. Suppose x and y both lie in every I_n , and toward a contradiction suppose $x \neq y$. Let $\varepsilon_0 = |x - y| > 0$. Since $|I_n| \rightarrow 0$, there is some n with $|I_n| < \varepsilon_0$. But $x, y \in I_n$ means $|x - y| \leq |I_n| < \varepsilon_0 = |x - y|$, which is absurd. Hence $x = y$. $\square$

So a nested sequence of intervals can “address” at most one point — and, geometrically, every point x on the line does arise this way:

Theorem 2.1.

Every real number x is the unique point contained in every interval of some nested sequence of closed intervals with rational endpoints.

Proof. Uniqueness is Proposition 2.1. For existence, pick any rational bracket $I_1 = [a_1, b_1]$ with $a_1 \leq x \leq b_1$, and repeatedly bisect: given $I_n = [a_n, b_n]$, let $m_n = \frac{a_n + b_n}{2}$ (rational, since $a_n, b_n \in \mathbb{Q}$), and set

$$I_{n+1} = \begin{cases} [a_n, m_n] & \text{if } x \leq m_n \\ [m_n, b_n] & \text{if } x > m_n. \end{cases}$$

Then $x \in I_{n+1} \subseteq I_n$ and $|I_{n+1}| = |I_n|/2$, so $|I_n| = |I_1|/2^{n-1} \rightarrow 0$. This is a nested sequence of closed intervals with rational endpoints containing x . $\square$

In this sense, every real number can be described — completely and unambiguously — by an infinite sequence of rational numbers (the endpoints a_n, b_n).

2.2 The Nested Interval Axiom

The construction above starts from a real number x and produces a nested sequence around it. What about the converse: if we are simply *handed* a nested sequence of closed intervals with rational endpoints, must there be a point in $\bigcap_n I_n$ at all?

For **open** intervals, the answer is no: $I_n = (0, \frac{1}{n})$ satisfies $I_1 \supseteq I_2 \supseteq \dots$ and $|I_n| = \frac{1}{n} \rightarrow 0$, yet no point lies in every I_n — any candidate $x > 0$ fails once $n > 1/x$, and $x = 0$ is excluded from every I_n by openness.

For **closed** intervals, this is far less obvious, and cannot be deduced from the algebraic properties of $\mathbb{Q}$ alone (after all, $\mathbb{Q}$ itself has a gap at $\sqrt{2}$). We take it as a basic postulate about the number line.

Axiom (Nested Interval Property)

If $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$ is a nested sequence of closed intervals with rational endpoints, then

$$\bigcap_{n \in \mathbb{N}} I_n \neq \emptyset,$$

i.e. there is a point x contained in every I_n .

This is a genuine **axiom**: a basic postulate we accept without proof, rather than a consequence of things we already know. Together with Proposition 2.1, it guarantees that no gaps exist on the number line — every nested sequence of rational brackets zeroes in on exactly one real number, no more and no less. This completeness of the number line is fundamental for everything that follows in this course; we will meet it again, in other guises, once we get to limits.

2.3 Decimal Expansions

Decimal expansions are simply a very concrete, systematic way of producing a nested sequence of intervals — one that subdivides by tens at every step.

Definition 2.2.

Let x be a point on the number line. Set

$$c_0 := \lfloor x \rfloor,$$

the greatest integer with $c_0 \leq x \leq c_0 + 1$, and $I_0 = [c_0, c_0 + 1]$, so $|I_0| = 1$.

Divide I_0 into ten equal parts using the points $c_0 + \frac{1}{10}, c_0 + \frac{2}{10}, \dots, c_0 + \frac{9}{10}$, and let $c_1 \in \{0, 1, \dots, 9\}$ be the digit with

$$c_0 + \frac{c_1}{10} \leq x \leq c_0 + \frac{c_1}{10} + \frac{1}{10},$$

and set I_1 to be this interval, so $|I_1| = \frac{1}{10}$.

Iterating, having chosen digits $c_1, \dots, c_{n-1} \in \{0, \dots, 9\}$, subdivide I_{n-1} into ten equal parts and let $c_n \in \{0, \dots, 9\}$ be the digit with

$$c_0 + \frac{c_1}{10} + \dots + \frac{c_n}{10^n} \leq x \leq c_0 + \frac{c_1}{10} + \dots + \frac{c_n}{10^n} + \frac{1}{10^n},$$

and let I_n be this interval, so $|I_n| = 10^{-n}$.

By construction $(I_n)_{n \geq 0}$ is exactly a nested sequence of closed intervals with rational endpoints in the sense of Definition 2.1: $I_0 \supseteq I_1 \supseteq I_2 \supseteq \dots$, each contains x , and $|I_n| = 10^{-n} \rightarrow 0$. We abbreviate this whole sequence by writing

$$x = c_0 + 0.c_1c_2c_3 \dots,$$

the **decimal expansion** of x — a notational shortcut for the nested intervals, not a new kind of number. (Careful when $c_0 < 0$: the digits after the point are still *added* to c_0 . For instance $x = -2.75$ has $c_0 = \lfloor -2.75 \rfloor = -3$ and $x = -3 + 0.25$, not $-3 - 0.25$.)

Combining Definition 2.2 with Proposition 2.1 and the Nested Interval Axiom:

- Every real number has an (infinite) decimal expansion.
- Conversely, every such expansion — i.e. every choice of $c_0 \in \mathbb{Z}$ and digits $c_1, c_2, \dots \in \{0, \dots, 9\}$ — is a nested sequence of closed intervals with rational endpoints, so by the axiom it defines a real number.

Decimal expansions are not always unique

Some real numbers have *two* decimal expansions: for instance

$$1.000 \dots = 0.999 \dots$$

Exercise 2.1. Why does $1.000 \dots = 0.999 \dots$? (Hint: what would $1 - 0.999 \dots$ have to be, if it were positive?) More generally, can you characterize *exactly* which real numbers have two decimal expansions?

Solution. If $1 - 0.999 \dots = \delta > 0$, then δ would have to be smaller than 10^{-n} for every n (since $0.\underbrace{9 \dots 9}_n = 1 - 10^{-n}$ already exceeds $1 - \delta$), forcing $\delta = 0$ — contradiction. So the two expansions name the same point.

In general, a real number has two decimal expansions exactly when it is a **terminating** decimal, i.e. of the form $m/10^n$ for some $m \in \mathbb{Z}$, $n \geq 0$: one expansion ends in all 0's, the other in all 9's. Every other real number has a unique decimal expansion.

2.4 Other Bases

Nothing in Definition 2.2 is special about the number 10 — we could subdivide each interval into any number $p > 1$ of equal parts instead of ten, producing digits $c_i \in \{0, 1, \dots, p-1\}$ and a **base- p (or p -ary) expansion** of x .

The case $p = 2$, the **binary expansion**, is exactly the bisection procedure from the proof of Theorem 2.1: at each step we only need to decide whether x lies in the lower or upper half of the current interval.

Example 2.1.

Let's find the first three binary digits of $x = 0.76$. Start with $I_0 = [0, 1]$, so $c_0 = 0$.

- Midpoint 0.5: since $0.76 \geq 0.5$, digit $c_1 = 1$ and $I_1 = [0.5, 1]$.
- Midpoint 0.75: since $0.76 \geq 0.75$, digit $c_2 = 1$ and $I_2 = [0.75, 1]$.
- Midpoint 0.875: since $0.76 < 0.875$, digit $c_3 = 0$ and $I_3 = [0.75, 0.875]$.

So $0.76 = 0.110\dots$ in base 2.

Exercise 2.2. Find the first three fractional digits of the base-2 expansion of $x = 0.3$, and the first three fractional digits of the base-3 expansion of $x = 0.76$.

2.5 Summary

- A **nested sequence of closed intervals with rational endpoints** (Definition 2.1) can contain at most one point (Proposition 2.1), and every real number arises as the unique point of some such sequence (Theorem 2.1).
- The converse — that every nested sequence of closed intervals with rational endpoints *does* contain a point — is not provable from what came before; it is the **Nested Interval Axiom**, the basic postulate that guarantees there are no gaps in the number line.
- **Decimal expansions** (Definition 2.2) are the special case of base-10 subdivision; every real number has one, though numbers of the form $m/10^n$ have exactly two (Exercise 2.1).
- The same construction works in any base $p > 1$, giving p -ary expansions; base 2 is exactly the bisection algorithm behind Theorem 2.1.